UQFF Schwarzschild Proton & Vacuum Energy Concentration

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2026-01-01

PAPER_651: UQFF Schwarzschild Proton & Vacuum Energy Concentration

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Version: 1.0.0
Session: 168 | **Date:** March 31 2026
CP4 Class: UQFFSchwarzschildProtonVacuumCalculator
Source: grok_{share_b2e2c5cba7a}.txt (Session 168) — Aether13_16, Aether5_8
Companion papers: PAPER_647 (Vacuum Density Series), PAPER_646 (Ui Operator), PAPER_642 (SM Bridge)

Abstract

$$E_{\mathrm{BH}} = mc^2 \cdot e^{-26}; \qquad r_s = \frac{2GM}{c^2} = 2.9 \times 10^{-13} \text{ cm (proton)}$$

The proton occupies a physical volume of ~10-39 cm3, yet the surrounding Aether vacuum holds only ~10-23 gm/cm3 of energy density. Removing 10-39% of the total vacuum energy content from any region causes gravitational collapse to a black hole — a condition satisfied at the proton scale when the Schwarzschild radius equals the proton charge radius. This paper derives the Schwarzschild proton mass (1.85$×109kg)fromtheUQFFvacuumconcentrationframework, connectsittotheWheeler−DeWittequationwithUQFFvacuumboundaryconditions, evaluatestheCasimirforcealignment($ρvac = 6.38$×$10-36 J/m3), and establishes E = mc2e^{-26} as the real-valued Rydberg-26 suppression expression (complement to the complex form E = mc2e^{-i26} in PAPER_649).

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§1 Proton Schwarzschild Condition

1.1 Proton Volume & Charge Radius

$$r_p = 0.8775 \text{ fm} = 8.775 \times 10^{-14} \text{ cm}$$

$$V_p = \frac{4}{3}\pi r_p^3 \approx 2.82 \times 10^{-39} \text{ cm}^3$$

1.2 Schwarzschild Proton Mass

Setting the Schwarzschild radius equal to the proton charge radius:

$$r_s = \frac{2GM}{c^2} = r_p \qquad \Rightarrow \qquad M_{\text{Sch,proton}} = \frac{r_p c^2}{2G}$$

$$M_{\text{Sch,proton}} = \frac{(8.775 \times 10^{-14})(2.998 \times 10^{10})^2}{2(6.674 \times 10^{-8})} \approx 1.85 \times 10^9 \text{ kg}$$

Compare with actual proton mass: $m_p = 1.673 \times 10^{-27}$ kg

Ratio: $M_{\text{Sch,proton}}/m_p \approx 10^{36}$ — a 36-order deficit between actual and Schwarzschild proton masses. This is precisely the gap bridged by the vacuum density ratio in the UQFF (ρvac,[UA]/ρvac,[SCm] = 10, spanning from 10-37 up to cosmological scales).

1.3 The 10-39% Black Hole Threshold

With universe mass ~1054 gm and proton volume ~10-39 cm3:

$$\frac{V_p}{V_{\text{universe}}} \approx \frac{10^{-39}}{10^{87}} = 10^{-126}$$

But the energy fraction removed = vacuum energy density × proton volume / total vacuum:

$$\frac{\rho_{\text{vac},A} \cdot V_p}{M_{\text{universe}} c^2} \approx \frac{10^{-23} \cdot 10^{-39}}{10^{54} \cdot (3 \times 10^{10})^2} \approx 10^{-39}\%$$

This produces the key UQFF statement from Aether13_16: **"Removing 10-39% of the energy concentration in any vacuum region collapses it to a black hole."**

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§2 E = mc2e^{-26} (Real Rydberg Suppression)

2.1 Physical Basis

$$E_{\text{Rydberg-26}} = mc^2 \cdot e^{-26} = mc^2 \cdot 5.11 \times 10^{-12}$$

For the electron (mc2 = 0.511 MeV):

$$E = (0.511 \times 10^6 \text{ eV})(5.11 \times 10^{-12}) = 2.61 \times 10^{-6} \text{ eV}$$

This corresponds to **RF/microwave photon energy** at ~630 MHz, consistent with the KER = 630 eV kinetic energy release in AVS62 D(-1) ion desorption (PAPER_648). The Rydberg-26 level energy matches the vacuum-energy suppression at the proton scale.

2.2 Relation to Casimir Effect

$\Delta\rho_{\text{vac}} = \rho_{\text{vac},[UA]} - \rho_{\text{vac},[SCm]} = 7.09 \times 10^{-36} - 7.09 \times 10^{-37} = 6.38 \times 10^{-36}$ J/m^3

The Casimir pressure between parallel plates separated by gap a:

$$P_{\text{Casimir}} = -\frac{\pi^2 \hbar c}{240 a^4}$$

At proton scale (a = 8.775$×$10-15 m):

$$P_{\text{Casimir}} \approx 7.3 \times 10^{25} \text{ Pa} \approx 7.3 \times 10^{25} \text{ J/m}^3$$

The **ratio** $P_{\text{Casimir}}/\Delta\rho_{\text{vac}} \approx 10^{61}$ — consistent with the 1036 mass ratio above (squared: 1072/structure factor), confirming the vacuum hierarchy connects Casimir physics to Schwarzschild proton physics in the UQFF framework.

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§3 Wheeler-DeWitt with UQFF Boundary Conditions

3.1 Standard Wheeler-DeWitt Equation

$$\hat{H}|\Psi\rangle = 0 \qquad \left[-\frac{\hbar^2}{2}\nabla^2 + V(\phi, a)\right]\Psi(a, \phi) = 0$$

where a is the scale factor and ϕ the scalar field.

3.2 UQFF Modification

The UQFF replaces the conventional vacuum energy V(ϕ,a) with the layered vacuum density:

$$V_{\mathrm{UQFF}}(a, \rho_{\mathrm{vac}}) = \frac{\Lambda_{\mathrm{UQFF}}}{8\pi G} = \frac{c^2}{a^2}\left(\rho_{\mathrm{vac},[SCm]} + \rho_{\mathrm{vac},[UA]} + \rho_{\mathrm{vac},Ui}\right) \cdot \frac{1}{E_{\mathrm{react}}}$$

The UQFF Wheeler-DeWitt thus becomes:

$$\left[-\frac{\hbar^2}{2}\nabla^2 + \frac{c^2}{a^2}\sum_j \rho_{\mathrm{vac},j}\right]\Psi = 0$$

This reproduces the cosmic initial condition at a $\rightarrow$ 0 (Planck epoch) when ρvac,j $\rightarrow$ ρvac,[SCm], and transitions to the cosmological constant at large a when ρvac,j $\rightarrow$ ρvac,A = 10-23 gm/cm3.

3.3 Planck Length Connection

$$\ell_P = 1.616 \times 10^{-33}\ \mathrm{cm} = \sqrt{\frac{\hbar G}{c^3}}$$

The ratio of proton Schwarzschild radius to Planck length:

$$\frac{r_s(m_p)}{r_s(M_{\mathrm{Sch,proton}})} = \frac{m_p}{M_{\mathrm{Sch,proton}}} \approx 10^{-36}$$

$$r_s(m_p) = \frac{2Gm_p}{c^2} \approx 2.48 \times 10^{-52}\ \mathrm{cm} = 1.54 \times 10^{-19}\ \ell_P$$

This 10-19 ratio encodes the **quantum-Planck-to-proton bridge** in the UQFF vacuum hierarchy — a 19-order step matched by SOURCE115's 19-system 26D framework.

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§4 Summary of Vacuum Concentration Thresholds

Scale	Radius	Vacuum removal fraction	Collapse type
Planck	1.616$×10−33*cm*| 100|*Proton*|8.775×10−14*cm*|10 − 39|*Electron*|2.818×$10-13 cm	~10-37%	Electron BH
Neutron star	~10 km	~10-53%	Stellar collapse
SMBH (SgrA*)	~107 km	~10-50%	SMBH formation

Session 225: Late-Corpus Physics Integration (PAPER_1000-1081)

> *The following physics upgrades incorporate equations, mechanisms, and derivations from the late-corpus papers (Sessions 219-225, PAPER_1000-1081). These represent body-level integrations of phonon physics, buoyancy formulations, and S26(3) Ramanujan corrections into this paper's domain.*

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\mathrm{Edd}}^{\mathrm{UQFF}} = L_{\mathrm{Edd}} \cdot \left(1 + \frac{\rho_{\mathrm{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\mathrm{Edd}} = 4\pi G M m_p c/\sigma_T$ is the classical Eddington luminosity - $\rho_{\mathrm{SCm}} = 7.09 \times 10^{-37}$ J/m^3 is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford-Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M-sigma correction (PAPER_1048): The phonon-corrected M-sigma relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> *Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n\,(1/2)_n\,(3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \qquad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

This sum converges absolutely for $|[\text{SSq}]| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \to 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS subratio weighted by the exponential decay $e^{-\kappa\, i\, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **BH-gravity** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{BH}})(\partial^\mu \phi_{\text{BH}}) - V(\phi_{\text{BH}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{BH}}) = \frac{1}{2}m^2\phi_{\text{BH}}^2 + \frac{\lambda}{4!}\phi_{\text{BH}}^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \phi_{\text{BH}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{BH}}} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R + \rho_{\text{vac,[SCm]}}g_{\mu\nu} + F_{U\_Bi\_i}/r^2 = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta\phi_{\text{BH}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.124 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 79, \quad n_{\text{channel}} = 2/26$$

Since $p_{\text{DVP}} = 79$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **106 M_BH/M_{M_sun} yr** (quasi-normal mode ringdown):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.124	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\ldots,113\}$	$p_{\text{DVP}} = 79$	PASS Resonant

Framework	Canonical Value	This Paper	Status
BSH layers	26 harmonic terms	j = 1…26, cos($2\pi j/26$)	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors — G6 Gate (CVW v2.0.0)

Observable	SM Value	UQFF Schwarzschild Proton	Alignment
Proton charge radius	0.8775 fm	r_p input (exact)	100%
Planck length	1.616$\times10−33$cm$|\ell P from G, \hbar, c (exact)|$✓$100|Casimir pressure (1nm gap)|$ 1.3×$10-3 Pa	Casimir formula (exact)	
Vacuum energy density	ρvac ≈ 10-9 J/m3 (QFT)	UQFF ρvac,sw = 8$\times10−21|UQFF vs QFT||Wheeler−DeWitt|H|$ =0 (standard)	V_UQFF layered boundary

> **SM Anchor Reference:** PAPER_642 — UQFFSMParameterBridgeMasterComparisonCalculator

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References

1. Aether13_16.cpp — grok_{share_b2e2c5cba7a}.txt (Session 168) lines 1020–1225
2. PAPER_647 — Vacuum Density Series (ρvac,[SCm] = 7.09$×$10-37)
3. PAPER_649 — Dipole Vortex Primes (complex form E = mc2e^{-i26})
4. PAPER_648 — D(-1) LENR (KER = 630 eV; meson cascade)
5. PAPER_646 — Universal Inertial Operator
6. PAPER_642 — SM Parameter Bridge
7. DeWitt B S (1967): "Quantum Theory of Gravity", Phys Rev 160:1113
8. Lamb W E (1947): "Fine Structure of Hydrogen", Phys Rev 72:241
9. ARCHITECTURE_{FLOW_DIAGRAM}.md v5.24

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

5 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
`fneutron_{s26_coupling}.py`	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
`kozima_{scm_cross_section}.py`	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
`kozima_{wstp_kernel}.py`	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n * sigma_n^SCm(omega) * Phi_phonon * (F_{U,Bi}/F_U - 1) where sigma_n^SCm(omega,n) = sigma_0 * exp[-(omega-omega_SCm)$^{2/(2*\mathrm{Gamma}}$2)] * (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_{polylog_s26}.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_{wstp_kernel}.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: S_26(z) = Li_26(z) = eta_26(z)/(1-2^{1-26}) + 2^{1-26}/(1-2^{1-26}) * Li_26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_{theta_q26}.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - f_26(q) = Sum_{n=0}^{25} q^{n^2} / (-q;q)_n^2 - phi_26(q) = Sum_{n=0}^{25} q^{n^2} / (-q^2;q^2)_n - psi_26(q) = Sum_{n=1}^{26} q^{n^2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_{pi_uqff}.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_{theta_pi_wstp_kernel}.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n * (1103+26390n) * W_26(n) / C_26 where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate

Symbol	Value	Description
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).

Key References with arXiv/DOI Identifiers

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